

Tribhuvan University
Institute of Science and Technology
School of Mathematical Sciences
Second Re-Assessment 2082

Subject: **Advanced Data Mining**

Full marks: **45**

Course No: **MDS 602**

Pass Marks: **22.5**

Level: **MDS / II Year / III Semester**

Time: **2 hrs**

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [5x3=15]

1. Differentiate between predictive and descriptive data mining tasks.
2. Differentiate between Probabilistic and Deterministic Classifiers.
3. What is the difference between K-means and K-means++.
4. How is Iterative Dichotomizer 3 (ID3) different from C4.5 algorithm?
5. How do Box-and-Whisker, Histograms, and Scatter plots serve as visual diagnostic indicators for outliers.

Group B [5x6= 30]

6. What is HDBSCAN ?

OR

Define Ensemble Learning techniques. Discuss the specific operational scenarios where ensemble methods significantly outperform individual base classifiers.

7. How Apriori Principle drastically reduces the search space during frequent itemset generation.

OR

What are different type of evaluation metrics used for classification models?

8. Define the metrics of Support, Confidence, and Lift in the context of Market Basket Analysis.
9. How can we build the Association based recommendation system?
10. Explain how the DBSCAN algorithm utilizes core, border, and noise point definitions to natively isolate global data outliers without manual contamination tuning.
